

Deep Residual Learning for Academic Document Processing

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Abstract

Deep neural networks have revolutionized document processing. In this paper, we present an end-to-end asynchronous mic

1. Introduction

Academic research papers from publishers such as IEEE and Springer present unique layout challenges. Traditional machi

1.1 Problem Formulation

Let D be an input academic document consisting of N pages. Each page contains textual glyphs, formulas, and structural bo

2. Mathematical Modeling and Attention Mechanism

The self-attention mechanism operates on query (Q), key (K), and value (V) matrices computed from input tokens:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

where d_k represents the dimensionality of key vectors. Multi-head attention extends this:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^O$$

2.1 Optimization and Loss Function

We optimize our objective using cross-entropy loss with AdamW optimizer:

$$\mathcal{L}_{CE} = -\sum_{i=1}^M y_i \log(\hat{y}_i) + \lambda \|\theta\|^2$$

where $\lambda = 0.01$ is the weight decay regularization parameter.

3. Experimental Results and Performance Analysis

Table 1 presents the comparative benchmark of translation models across standard academic datasets:

Model Architecture	Parameters	BLEU Score	Latency (ms)	Throughput (tok/s)
CTranslate2 NLLB-200	600M	34.2	243.2	128.5
Qwen2.5-7B Instruct	7.6B	41.8	412.0	85.2
LLaMA-3.1-8B Instant	8.0B	40.5	294.1	142.0
Gemini 2.5 Flash Large		43.1	680.5	110.4

3.1 Ablation Study on Glossary Injection

When domain-specific glossary is injected into system prompts, the specialized terminology accuracy increases from 68.4%

4. Conclusion and Future Directions

In this work, we demonstrated that hierarchical parent-child chunking combined with Markdown-intermediate representation

References

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